BA ZC581 :
Applied Data Science
Module: 10

BITS Pilani

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J A Gokhale.

Disclaimer

- The slides presented here are obtained / extracted from the authors of the prescribed books and from various other subject contributors.
- I hereby thank and acknowledge all the contributors for their material and inputs.
- I have added and modified a few slides to suit the requirements of the course.

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Module 10:
EDA: Visual Exploration

Module 10 Topics

Topic 1: Introduction to EDA

Topic 2: Plotting Libraries- Seaborn & Matplotlib

Topic 3: Histograms, Boxplots

Topic 4: Scatterplots, Pairplots

Topic 5: Demos

Ref: T1 , Ch 4, 16

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Topic 1: Introduction to EDA

Exploratory Data Analysis (EDA)

- EDA is the process of analyzing datasets to summarize their main characteristics, often using visual methods.
- Purpose:
  - Understand data structure and patterns
  - Summarization gives us some information that helps us in building the model
  - Detect anomalies and outliers
  - Form hypotheses and guide further analysis
  - We get different perspectives of data from visualizing
- Key tools:
  - Statistics, plots, and domain knowledge

EDA

A process used to understand the data better even before preparing the model.

EDA is primarily used to see what data can reveal beyond the formal modeling or hypothesis testing task and provides a provides a better understanding of data set variables and the relationships between them.

It can also help determine if the statistical techniques one is considering for data analysis are appropriate.

It can help identify obvious errors, as well as better understand patterns within the data, detect outliers or anomalous events, find interesting relations among the variables. These provide insights about the data

Analyze the data using

- Statistical summarization
- Visual representation

EDA

EDA

Basic Data Cleaning	Data Summarization	Distributions in Data	Relationships in Data	Spot issues in Data
• Missing values • Type conversion • Encoding	• Aggregation • Grouping • Binning	• Histograms • Box plots	• Scatter plots • Pair plots	• Outliers • Missing data

Common Steps in EDA

- Data Collection & Loading
- Data Cleaning:
  - Handle missing values
  - Remove duplicates
  - Correct data types
- Data Profiling:
  - Summary statistics (mean, med
    dev)
  - Distribution checks
- Visualization:
  - Histograms, bar charts, scatter plots, box plots

Quick Understanding of Data.

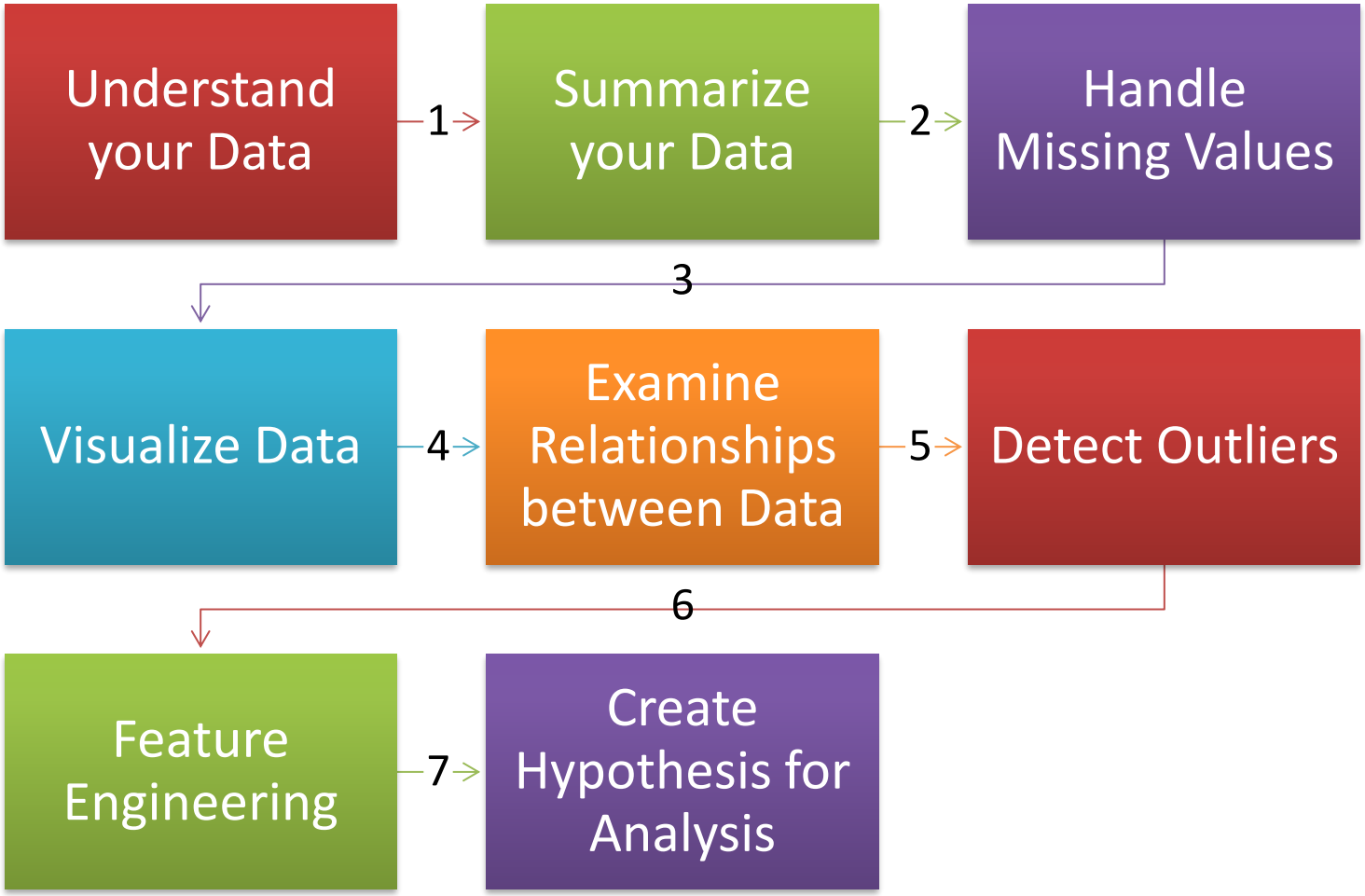
Visual Inspection

Basic insights

The EDA Pipeline

1. Understand your Data
2. Summarize your Data
3. Handle Missing Values
4. Visualize Data
5. Examine Relationships between Data
6. Detect Outliers
7. Feature Engineering
8. Create Hypothesis for Analysis

The EDA pipeline



Need for EDA

- Gain an understanding of the data
- Examine various perspectives of the data
- Identify core features of the data
- Estimate underlying structure of data
- Identify irregularities like outliers and missed relations
- Clear and ascertain core assumptions on the data

Basic Types of EDA

Mathematical / Statistical

Visual / Graphical

Combination of both

Types of EDA Techniques

Univariate Analysis:

Focus on one variable at a time.

Provides statistical summary of every variable

Tools: histograms, box plots, frequency tables

Bivariate Analysis:

Explore relationships between two variables

Estimate relationship between each variable to the target variable

Tools: scatter plots, correlation matrices, cross-tabulations

Multivariate Analysis:

Examine interactions among multiple variables

Tools: pair plots, heatmaps, dimensionality reduction (PCA)

Dimensionality Reduction:

Identifies fields that exhibit maximum variance

Reduces the number of variables for analysis (and hence the data volumes)

What is done in EDA?

- Estimate, Evaluate & Visualize:
  - Summary Statistics
    - Mean, Median, Mode
  - Distributions & Spread Measurements
    - Histograms, Distribution plots
    - Box plots, Skewness
  - Relationships
    - Correlations
    - Co-variance
  - Hypothesis Testing
  - Trends

Tools and libraries used in EDA

Python Libraries:	pandas for data manipulation matplotlib and seaborn for visualization plotly for interactive plots
Notebook Platforms:	Jupyter, Google Colab
BI Tools:	Power BI, Tableau (for visual EDA)

Ex: Titanic Data set

- Goals:
  - Understand survival patterns
  - Explore age, gender, class distributions
- Insights:
  - Higher survival rate for women and children
  - Class and age strongly influence outcomes

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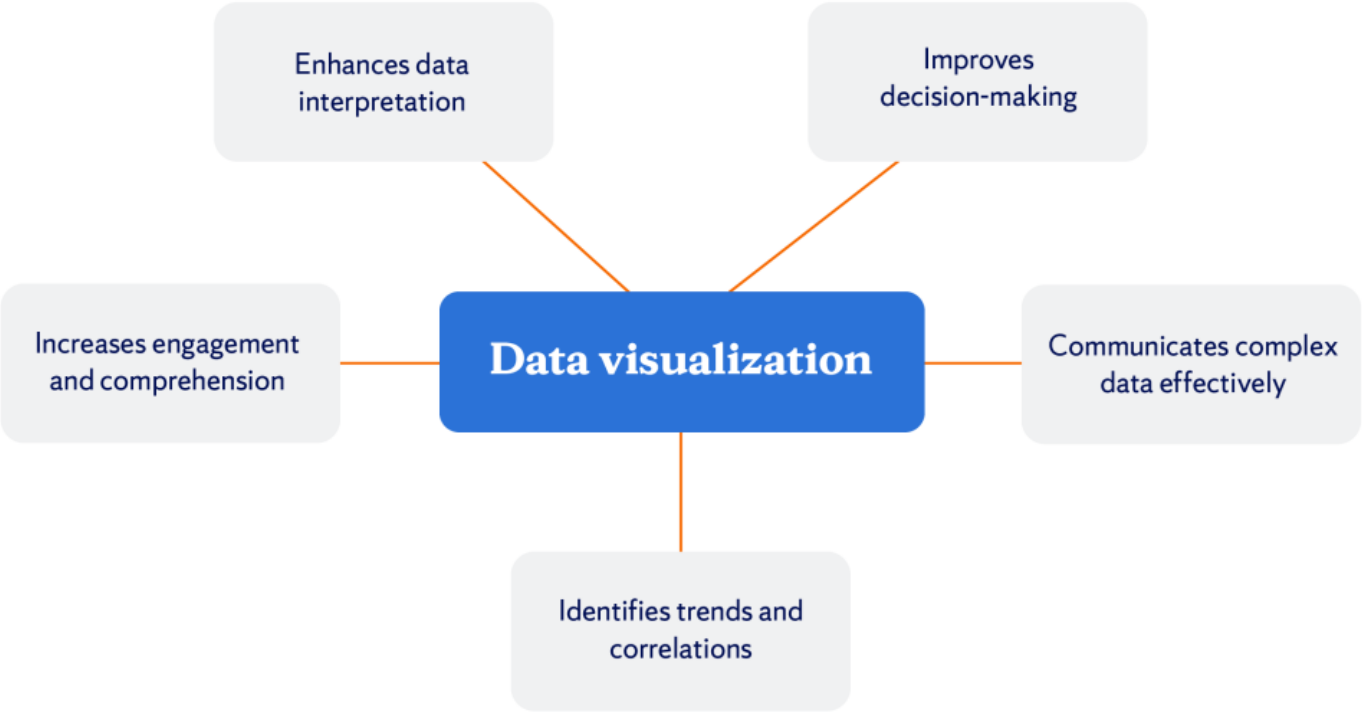
Topic 2: Plotting Libraries- Seaborn & Matplotlib

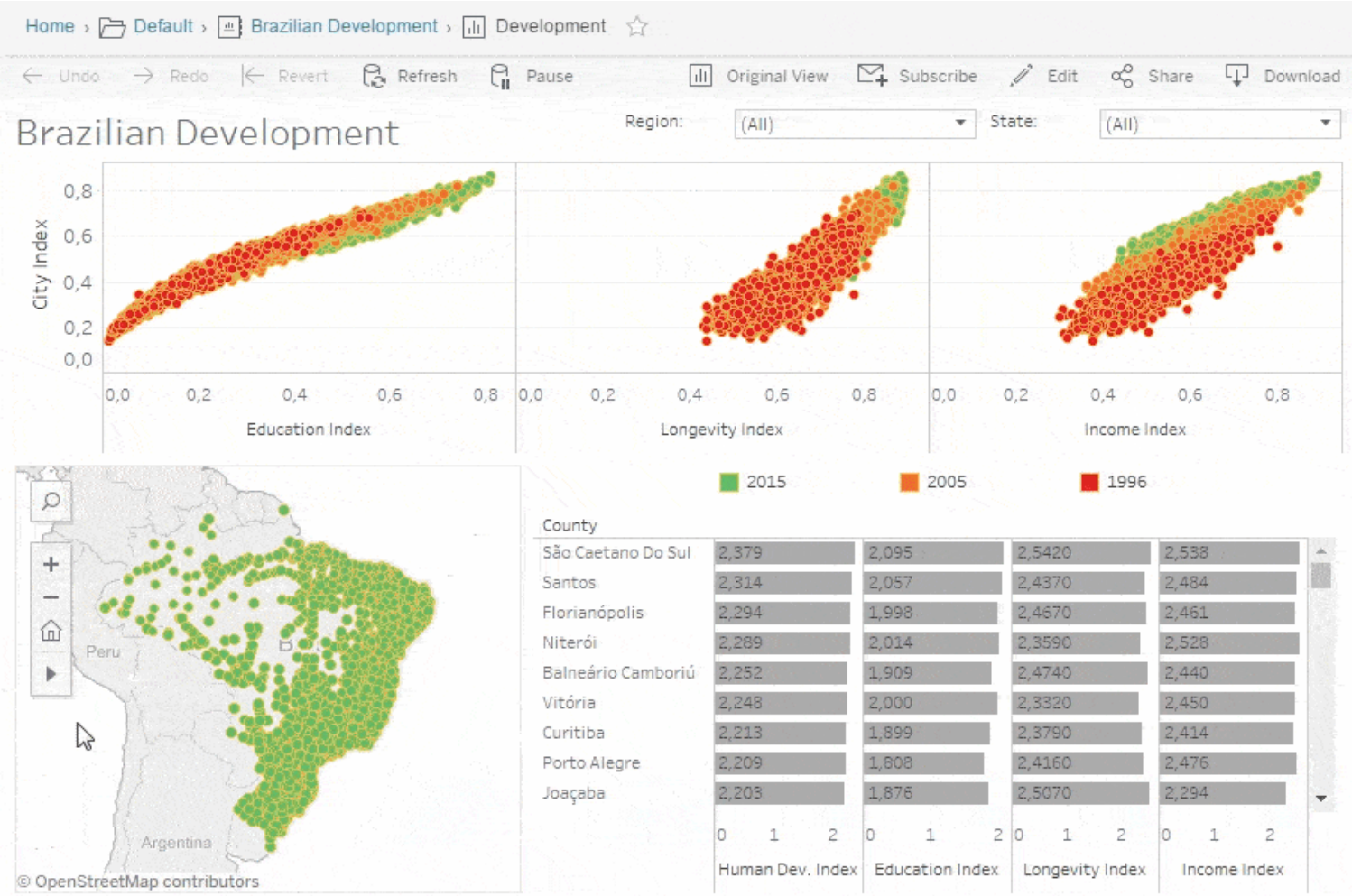
Data Visualization

- Data visualization is the practice of transforming complex data and information into visual formats like charts, graphs, maps, and dashboards, making patterns, trends, outliers, and insights easily understandable for faster, smarter decision-making, even for non-technical audiences.
- It translates raw numbers into accessible graphics, revealing stories within the data that would be hard to see in text / spreadsheets.
- A pictorial representation of data for easy understanding
- An Executive reporting mechanism

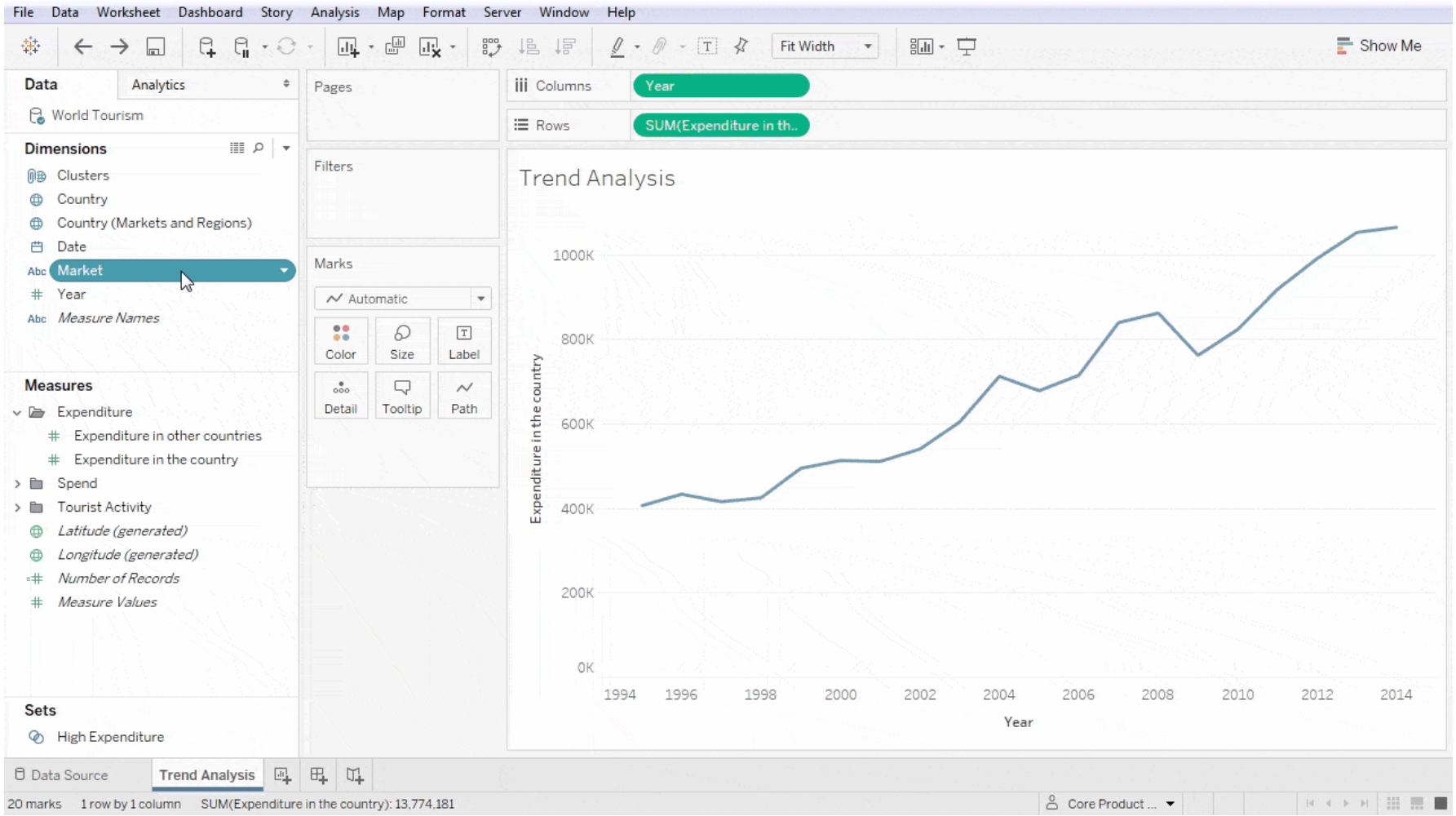
Data Visualization



An Example: Dashboard using Tableau



Another Example: Trend Analysis using Tableau



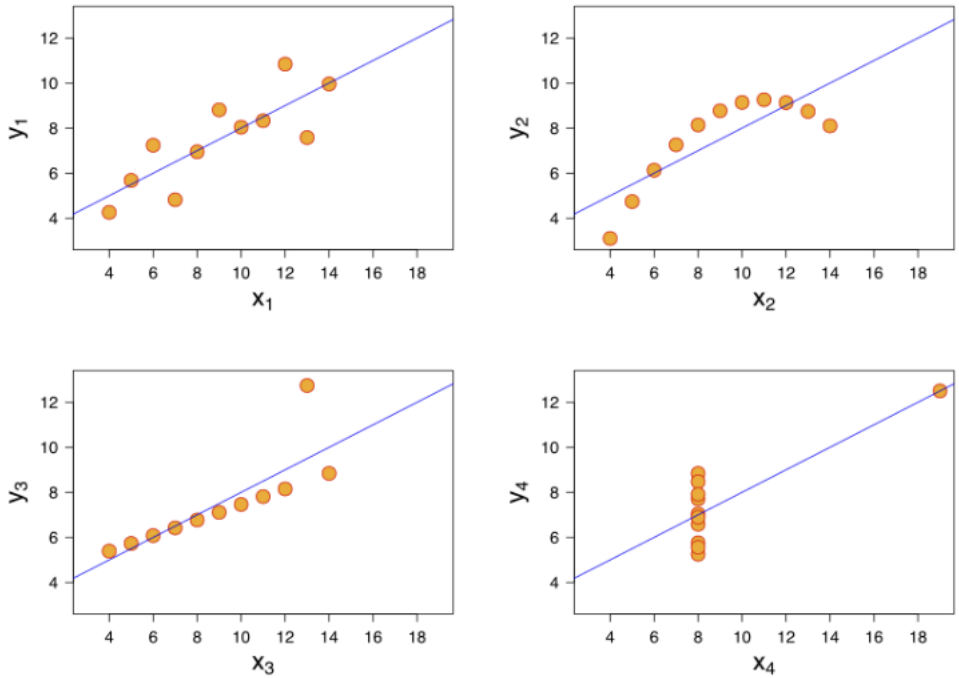
Example

- 4 Data sets which are almost identical (same descriptive statistics like mean, variance and correlation)

I		II		III		IV	
10	8.04	10	9.14	10	7.46	8	6.58
8	6.95	8	8.14	8	6.77	8	5.76
13	7.58	13	8.74	13	12.74	8	7.71
9	8.81	9	8.77	9	7.11	8	8.84
11	8.33	11	9.26	11	7.81	8	8.47
14	9.96	14	8.1	14	8.84	8	7.04
6	7.24	6	6.13	6	6.08	8	5.25
4	4.26	4	3.1	4	5.39	19	12.5
12	10.84	12	9.13	12	8.15	8	5.56
7	4.82	7	7.26	7	6.42	8	7.91
5	5.68	5	4.74	5	5.73	8	6.89

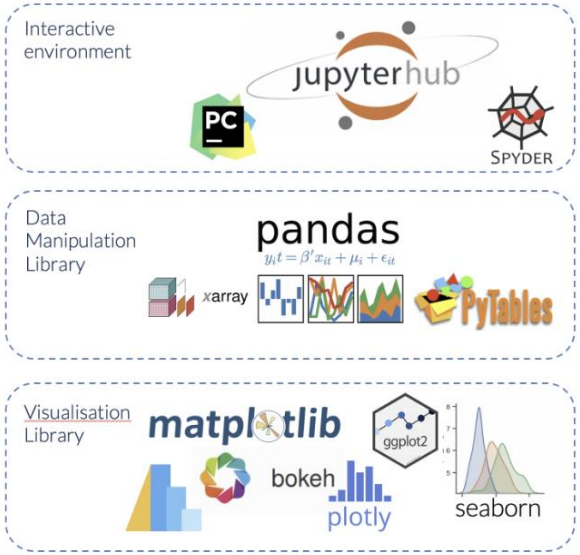
Example: Plotting the data …

- Datasets are unique…. but the plots???



Data Visualization Tools

- Data Visualization Tools
  - Power BI, Tableau, Qlik
- Cloud based
  - Google Data Studio, Google charts, Amazon QuickSight
- Programming Language Libr
  - Python libraries
  - R
  - Java
  - Javascript
  - C++



Python Libraries

- Python supports a rich set of libraries, mostly built on top of matplotlib.
  - **Matplotlib**: The foundational and most popular 2D plotting library, offering extensive control over plot elements. Many other libraries are built on top of it.
  - **Seaborn**: A high-level interface built on Matplotlib, ideal for creating attractive and informative statistical graphics with less code.
  - **Plotly**: A library for creating interactive, web-based visualizations with support for 2D and 3D charts, ideal for dashboards and online applications.
  - **Bokeh**: Focuses on high-performance interactive plots and dashboards for modern web browsers, with the ability to handle large or streaming datasets.

matplotlib

- A powerful, open-source Python library for creating static, animated, and interactive visualizations.
- Industry-standard for data visualization in Python.
- Highly customizable and integrates smoothly with NumPy, Pandas, and Jupyter.
- Wide range of plots: line, bar, scatter, histogram, pie, heatmaps.
- Fine-grained control over colors, styles, axes, annotations, and layouts.
- Supports multiple output formats (PNG, SVG, PDF).
- Object-oriented API for complex, multi-subplot figures.

matplotlib

- Typical workflow:
  1. Import: import matplotlib.pyplot as plt
  2. Prepare data: x-values, y-values etc
  3. Plot: plt.plot(x, y)
  4. Customize: add labels, title, legend
  5. Display: plt.show()

python

```
import matplotlib.pyplot as plt
x=[1,2,3]
y=[2,4,1]
plt.plot(x,y)
plt.title("Simple Line Plot")
plt.xlabel("X-axis")
plt.ylabel("Y-axis")
plt.show()
```

seaborn

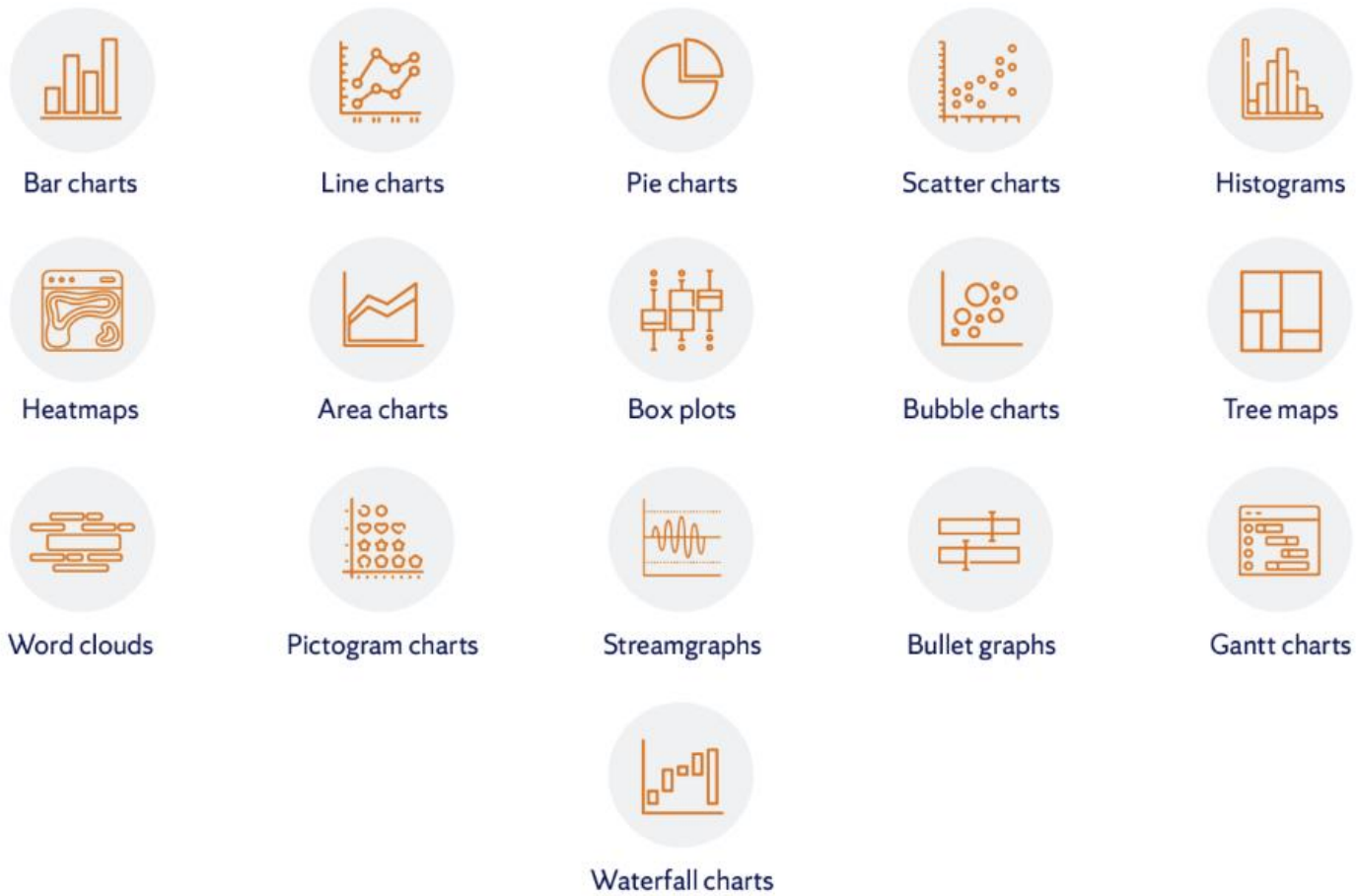
- A Python library for statistical data visualization, built on top of Matplotlib and tightly integrated with Pandas.
- Produces beautiful, publication-ready graphics (business use) with minimal code.
- Offers a declarative, dataset-oriented API, letting you focus on meaning rather than styling.
- High-level plot types: relational, categorical, distribution, regression, matrix, and multi-plot grids.
- Automatic statistical aggregation and semantic mapping (hue, size, style).
- Themes & aesthetics: built-in styles, color palettes, and easy customization.

seaborn

- Typical workflow:
  1. Import: import seaborn as sns
  2. Set theme: sns.set_theme()
  3. Load data (DataFrame)
  4. Plot: sns.relplot(...) / sns.histplot(...) etc.
  5. Refine: add and customize titles, labels, palettes
  6. Display: plt.show()

```
import seaborn as sns
sns.set_theme()
tips = sns.load_dataset("tips")
sns.relplot(data=tips, x="total_bill",
            y="tip", hue="smoker",
            col="time")
```

Data Visualization Types

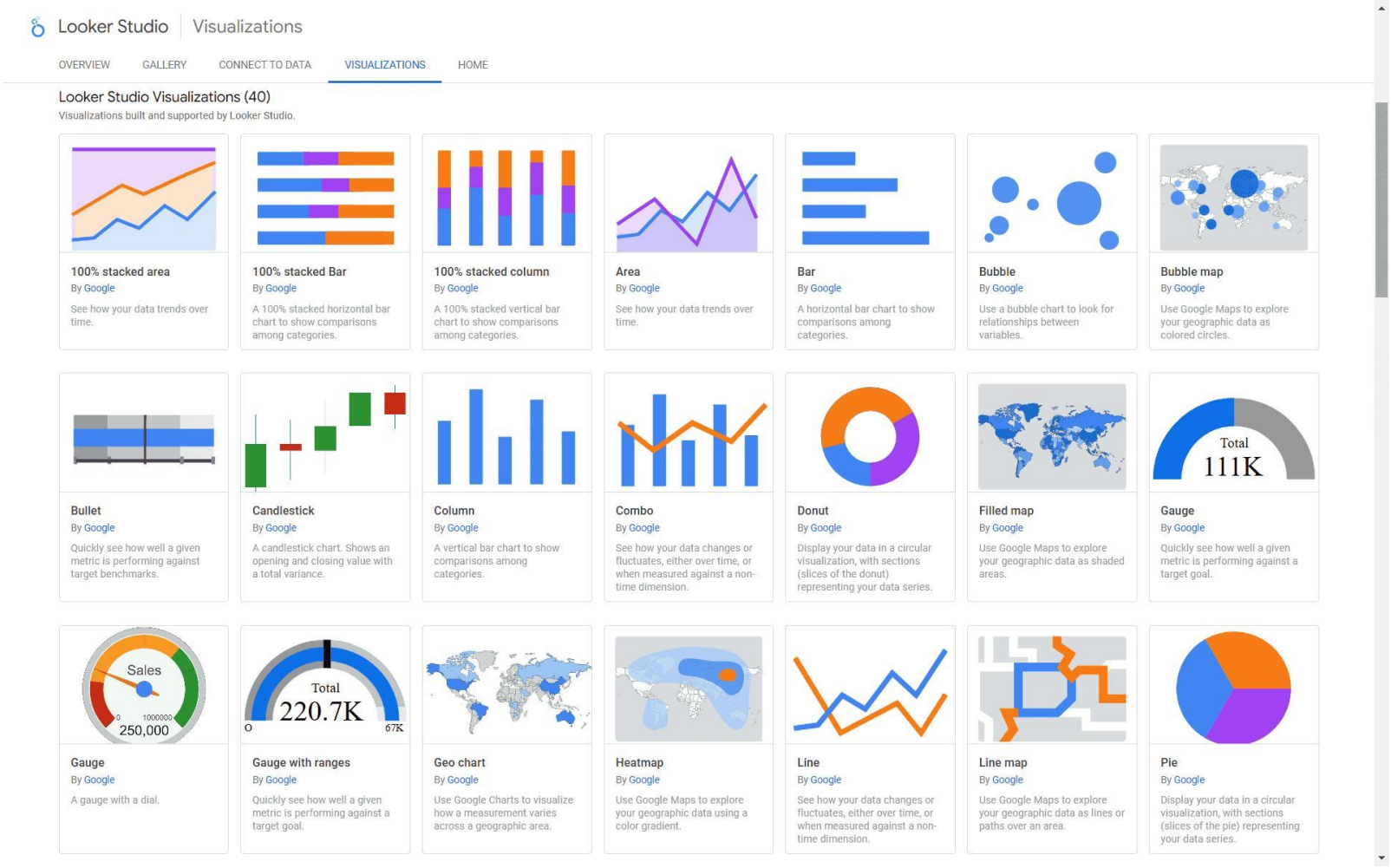


MS PowerBi

Bar
Column
Stacked bar / column
Line
Area
Multi / Combo
Ribbon
Waterfall
Funnel
Scatter / Bubble chart
Pie / Donut
Treeview
Violinplots

Maps
Gauge
Card / multicard
KPI
Table / matrix
Slicer
Key influencers
De-composition tree
Spider / Radar
Box-plot
Heatmaps
Pictograms

Looker Studio



Looker Studio

Common Plot Types ..

Bar charts – Compare categories by displaying rectangular bars, where the length represents the value.
Line charts – Show trends over time, with data points connected by lines.
Pie charts – Represent proportions of a whole, with each slice corresponding to a category's percentage.
Donut charts – Represents proportions of a whole like a Pie chart.
Scatter charts – Display relationships between two variables with data points on a graph.
Histograms – Visualize the distribution of numerical data by dividing it into intervals (bins).
Heatmaps – Use color to represent values in a matrix, identifying patterns and correlations.
Area charts – Similar to line charts, but with filled areas beneath the line to show the magnitude of change over time.
Box plots – Summarize data distributions by showing quartiles, outliers, and the median.
Bubble charts – Show the relationship between three variables using circles whose size and color provide extra data.
Tree maps – Display hierarchical data with nested rectangles, where each rectangle's size represents a value.
Word clouds – Visualize the frequency of words in text, with more common words appearing larger.
Pictogram charts – Use icons or symbols to represent data values in a visually appealing way.
Streamgraphs – Show how data composition changes over time with stacked areas.
Bullet graphs – Represent progress toward a goal, with markers indicating benchmarks.
Gantt charts – Track project timelines and tasks with horizontal bars indicating duration.
Waterfall charts – Display the cumulative impact of sequential data points, often used in financial analysis.
Funnel charts – Displays data in decreasing stages through a process
Ribbon charts – Display Ranking and contributions of data over time
Gauge chart – Displays a single value against a predefined scale as a needle on a dial
Maps – Displays data overlayed on a geographical map; used when location details need to be presented
KPI – Displays a KPI measure for quick understanding

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Topic 3: Histograms & Boxplots

Histograms

- Histogram: A graphical display showing the distribution of numerical data, grouping continuous values into "bins" or intervals and representing the count (frequency) of data points in each bin with the height of a vertical bar.
- Depicts the frequency distribution
- Bars usually touch each other to depict continuous intervals
- Reveals the shape, spread and central tendencies of the data
- Use histograms to understand the center of the data
- Univariate plot

Histograms

- How to generate?
  - Collect Data: Get a set of continuous numerical data (e.g., heights, temperatures, response times).
  - Define Bins: Divide the total range of your data into equal-sized intervals (equi-width bins).
    - Bin width = range / # of bins
  - Count Frequencies: Count how many data points fall into each bin.
  - Plot Bars: Draw bars for each bin; the x-axis shows the bin's value range, and the y-axis shows the frequency count.

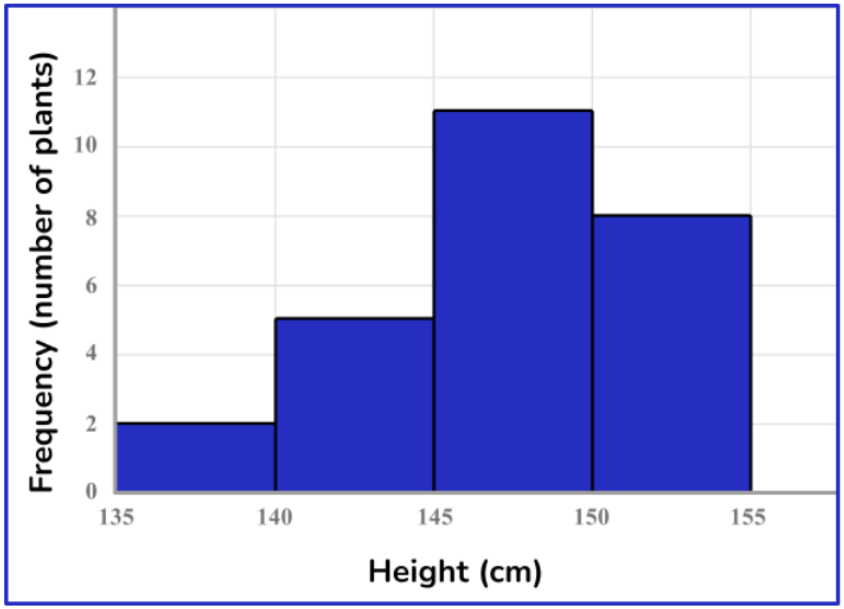
Example

Heights of plants in a garden

Height, x cm	Frequency
$135 \leq x < 140$	2
$140 \leq x < 145$	5
$145 \leq x < 150$	11
$150 \leq x < 155$	8

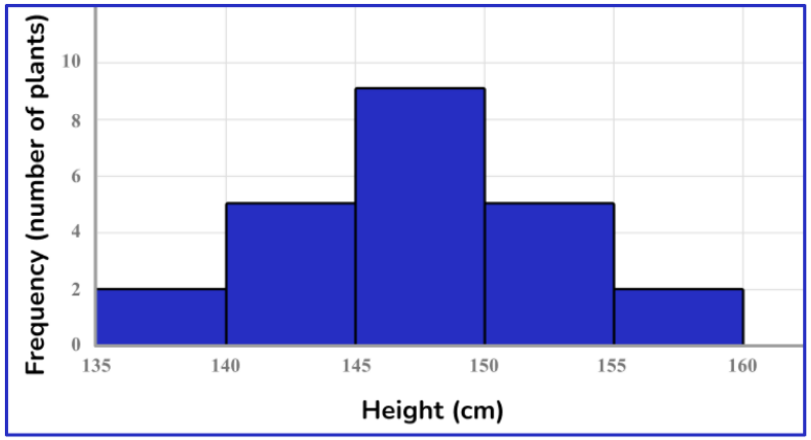
The bin size is 5 – each group has a range of 5. The values given for the beginning and end of the bin will be used on the x-axis. There are 4 bins – there are 4 frequencies. Each bin will be drawn as a different bar, so this graph will have 4 bars.
The height of each bar is the frequency, which is labeled on the y-axis.

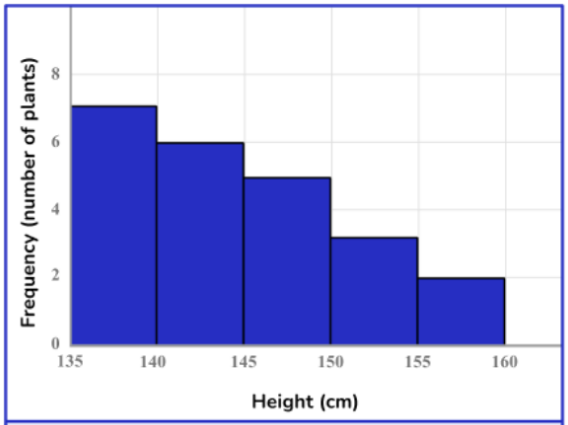
histogram



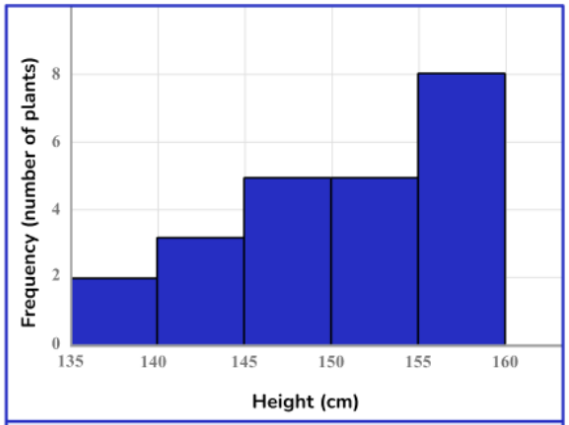
Other Distributions

NORMAL distribution





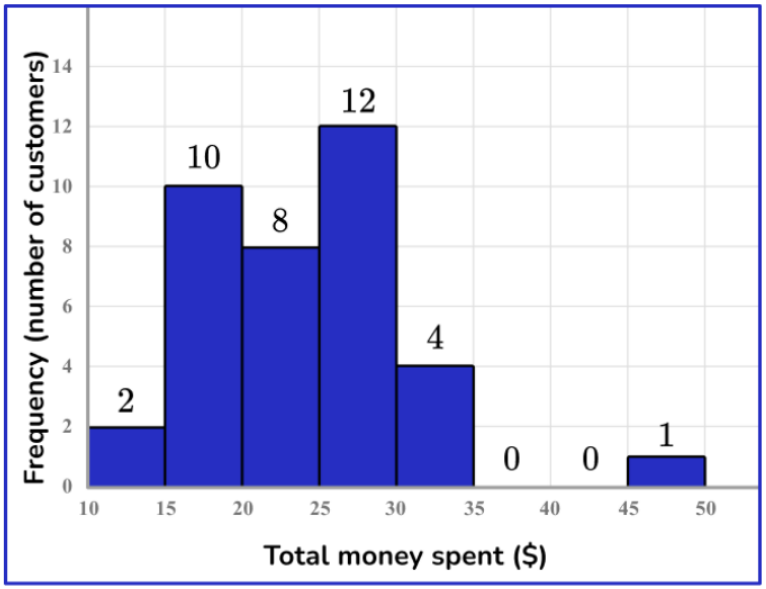
RIGHT SKEWED



LEFT SKEWED

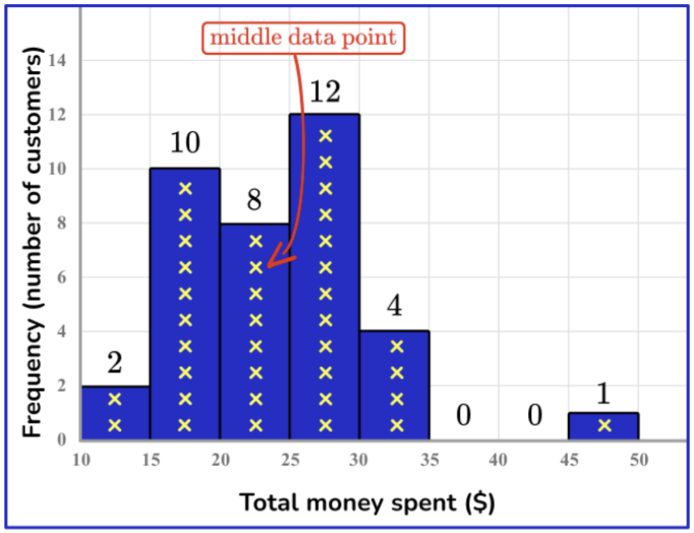
Exercise: Interpreting histograms

- Consider the following histogram.
- State the bin where the median lies and describe the center and spread of the data within context.



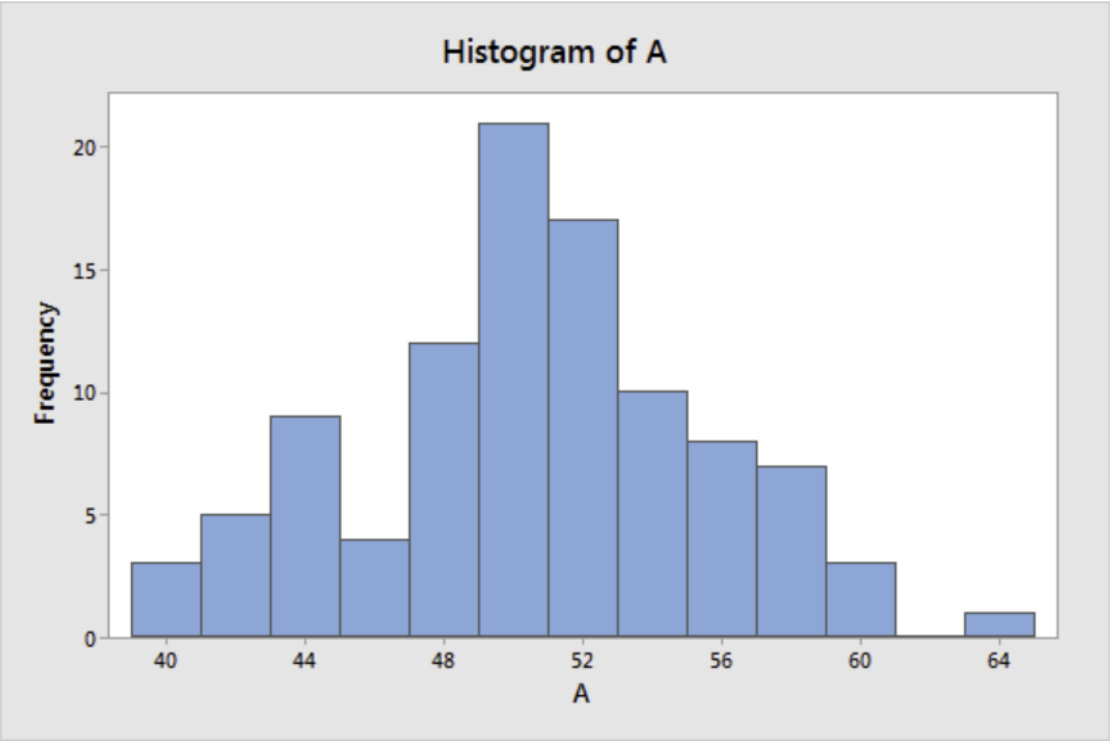
Exercise:

- Find the bin that has the median:
  - Start with the lowest and highest bins, cross across points in the bins.
- Most of the customers spent between $15 and $30.
- There was one customer who spent from $45 – $50, which seems to be an outlier.



Example Histogram

In the histogram below, you can see that it is a unimodal distribution because it has only one peak near 50. Most values in the dataset will be close to 50, and values further away are rarer. The distribution is roughly symmetric and the values fall between approximately 40 and 64.

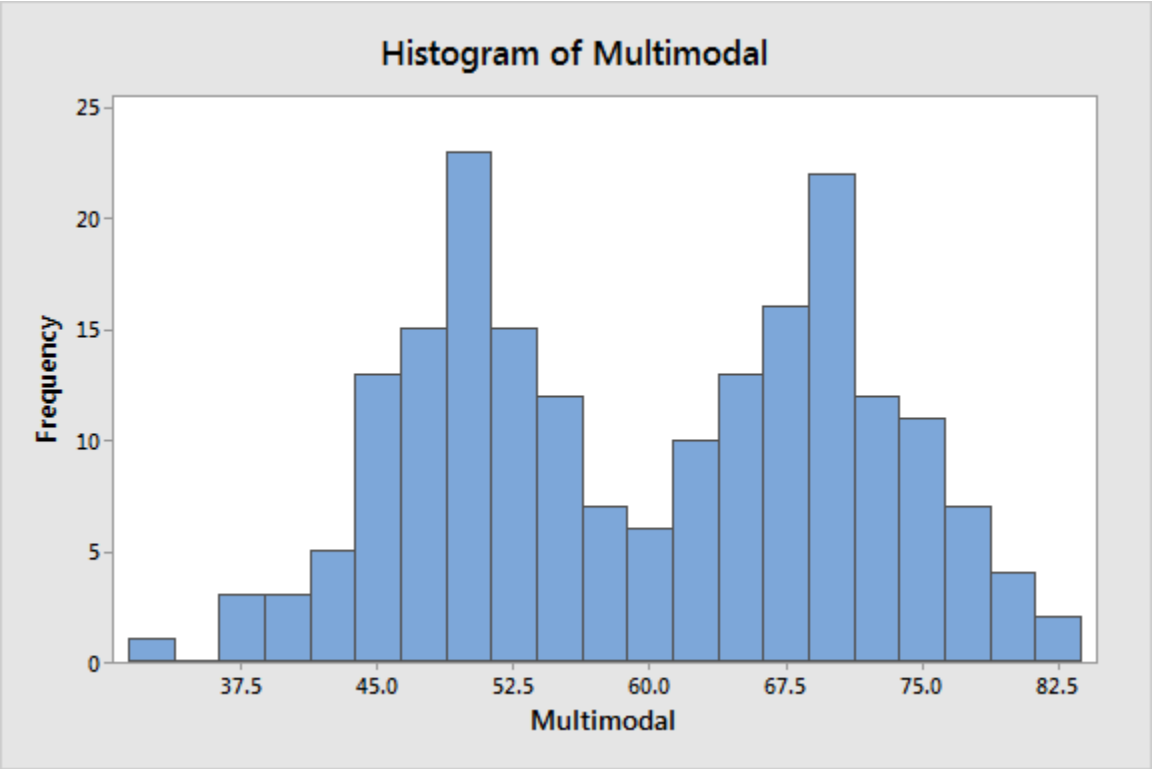


Histogram of a multi modal distribution

Descriptive stats:

Mean = 59.735; Median= 60.612; Std dev = 11.513

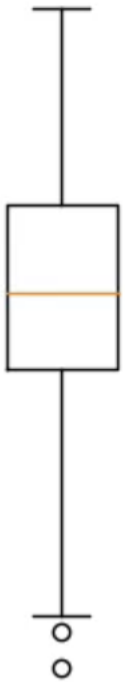
Plotting reveals multi modal distribution.



Box plots

- A box plot gives a compact visual summary of how a numerical variable is distributed.
- It highlights the center, spread, and unusual values in a clean, standardized way.
- What it shows:
  - Median (Q2)
    - Middle value of the data, line inside the box
  - Quartiles (Q1 and Q3)
    - Q1 (25th percentile): bottom of the box
    - Q3 (75th percentile): top of the box
    - Box represents IQR (Q3-Q1)
  - Whiskers
    - Lines extending the box; show minimum and maximum values within range
  - Outliers
    - Points outside whiskers, represent unusually low or high vues

Box Plots indicate..



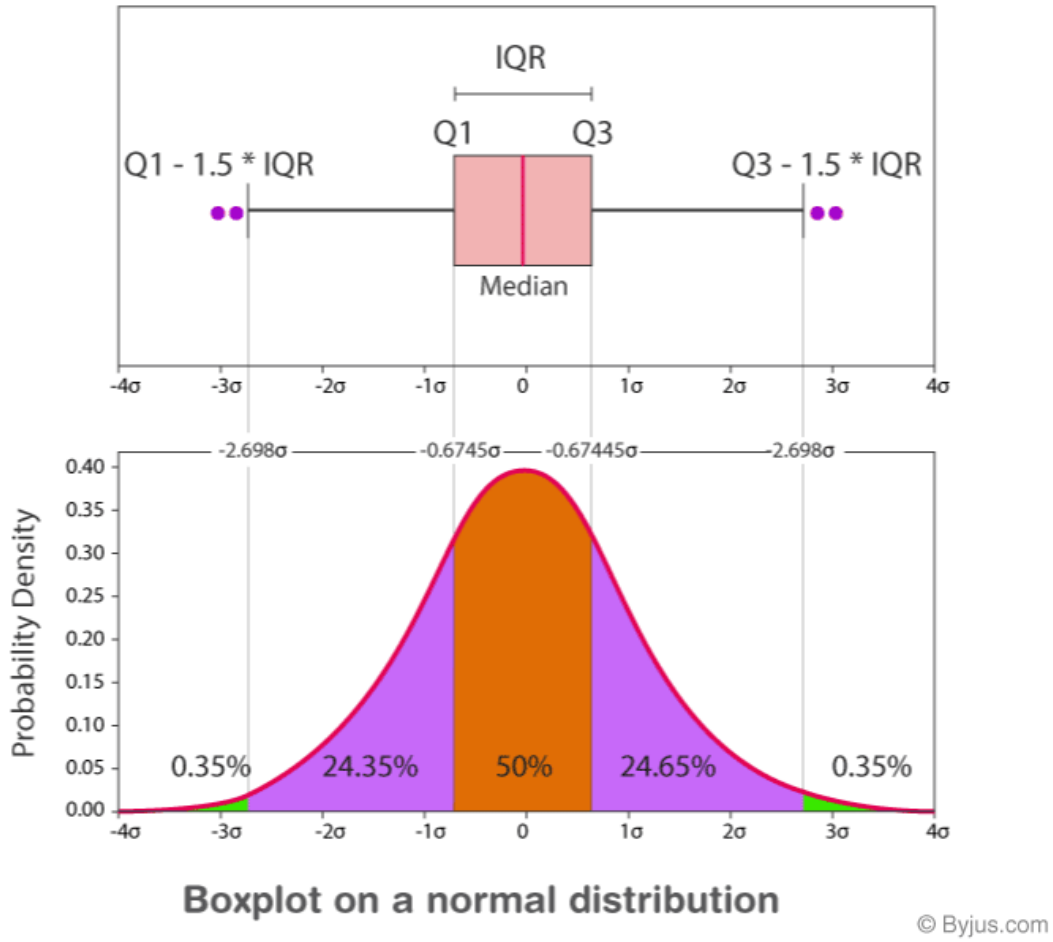
Spread: How wide the box is (variability).

Skewness: If the median is closer to one side.

Outliers: Any extreme values.

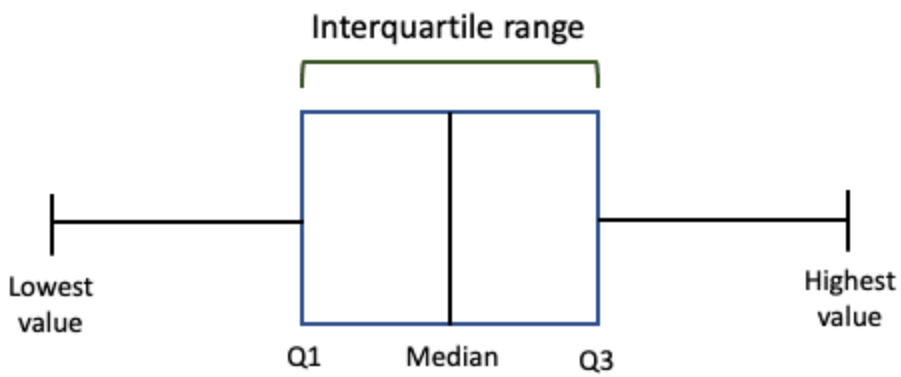
Comparison: Multiple box plots help compare distributions across groups.

Boxplot on a Normal Distribution



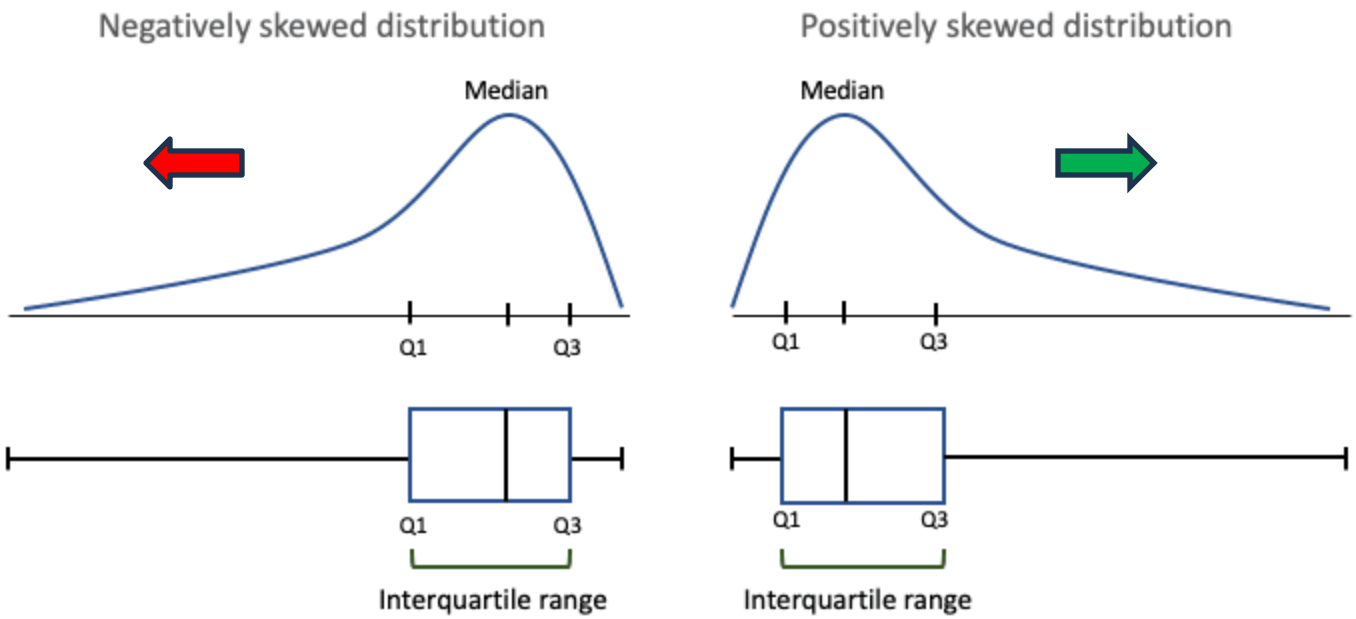
Box plot to visualize IQR

- IQR can give you an overview of where most of your values lie and how clustered they are.
- It is a good measure for data with skew
- The IQR is also useful for datasets with outliers. Because it's based on the middle half of the distribution, it's less influenced by extreme values



Box plot

- The placement of the box tells you the direction of the skew. A box that's much closer to the right side means you have a negatively skewed distribution, and a box closer to the left side tells you that you have a positively skewed distribution.



Example: IQR Computation

Step 1: Order your values from low to high.

48 52 57 64 72 76 77 81 85 88

Step 2: Locate the median, and then separate the values below it from the values above it.

Median

48 52 57 64 72 76 77 81 85 88

First half Second half

Step 3: Find Q1 and Q3.

Q1 Q3

48 52 57 64 72 76 77 81 85 88

First half Second half

Step 4: Calculate the interquartile range.

$$IQR = Q3 - Q1$$
$$IQR = 81 - 57 = 24$$

What if we now add values 32 and 320 ?

Outliers & IQR

- All values less than Q1 – 1.5(IQR) are probably too small
- All values greater than Q3+1.5(IQR) are probably too large

An example

- The wire cut machine's recorded time between failures is shown below (50 samples in increasing order):

2	22	32	39	46	56	76	79	88	93
3	24	33	44	46	66	77	79	89	99
5	24	34	45	47	67	77	86	89	99
9	26	37	45	55	67	78	86	89	99
21	31	39	46	56	75	78	87	90	102

- Mean, median and mode of time between failures:
  - Mean = 57.64; Median = 56; Mode = 46,89 and 99
  - Safe to assume machine will fail after 57 hours

Example, contd..

- Mode has 3 values:
  - First likely hood of failure after 46 hours; next support at 89 hours; finally at 99 hours!
- What time 10% and 90% of wire cut machines will fail?
  - Data is organised in increasing order.
  - P10 = 31 hours
  - P90 = 90 hours
  - By 31 hours 10% of wire cuts will fail & 90% of wire cuts will fail by around 90 hours

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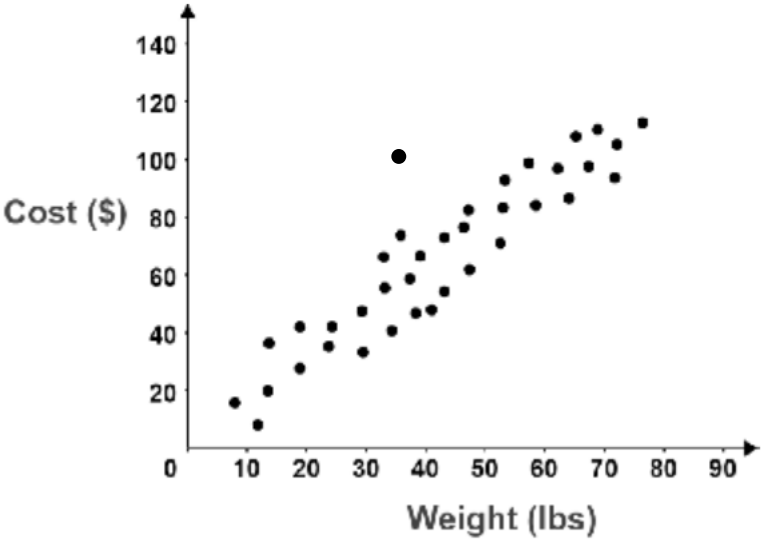
Topic 4: Scatter plots & Pair plots

Scatter plots

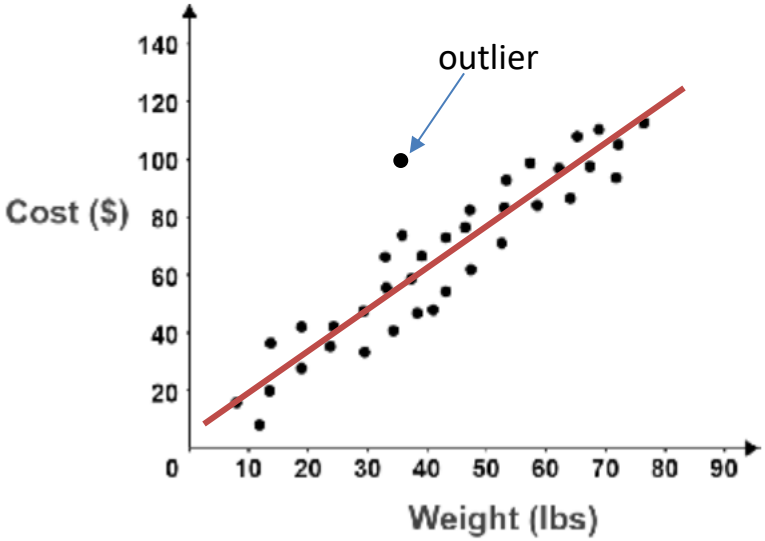
- A scatter plot (aka scatter chart, scatter graph) uses dots to represent values for two different **numeric** variables. (bi-variate graph)
- The position of each dot along the horizontal and vertical axis indicates values for an individual data point.
- Scatter plots are used to observe relationships between variables (correlation between variables)
- Help to visualize clusters of similar values
- Can identify outliers that do not follow any pattern
- Can help identify trends

Scatter Plot

Shows the correlation between two variables



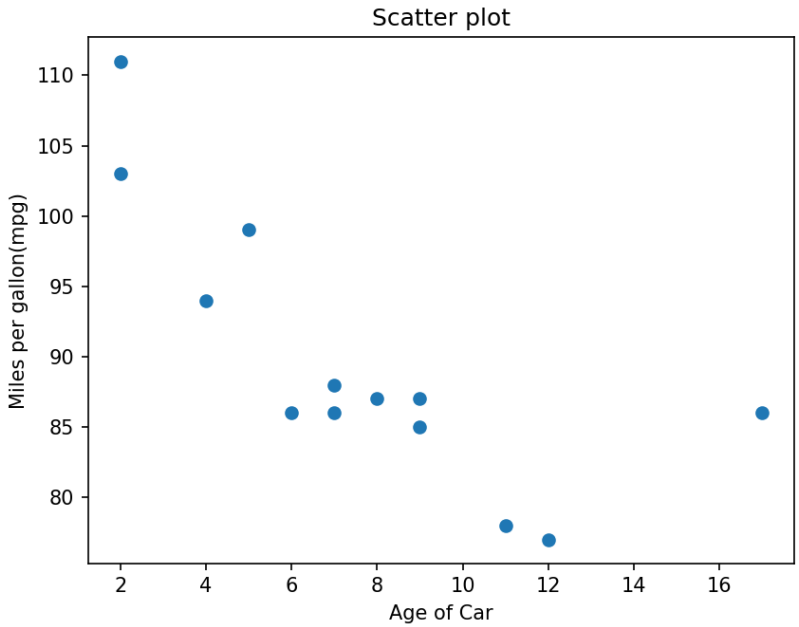
Weight vs. Cost($) scatter plot



Weight and cost are positively correlated

Python ..

```
# Basic Scatter plot trial program

import matplotlib.pyplot as plt

# Data
age_car = [5,7,8,7,2,17,2,9,4,11,12,9,6]
mpg = [99,86,87,88,111,86,103,87,94,78,77,85,86]

# draw scatter plot
plt.scatter(age_car, mpg)
plt.title("Scatter plot")
plt.xlabel("Age of Car")
plt.ylabel("Miles per gallon(mpg)")

plt.show()
```



Pair plots

- A pair plot is a **multi-variable visualization** that shows relationships between **every pair of numerical features** in a dataset.
- It's one of the most powerful quick-exploration tools in data analysis
- It is essentially a grid of scatter plots; creates scatter plots for every pair of numerical variables
- Creates histograms or KDE plots on the diagonal to show each variable's distribution
- Optionally colors points by a categorical variable (great for comparing groups)

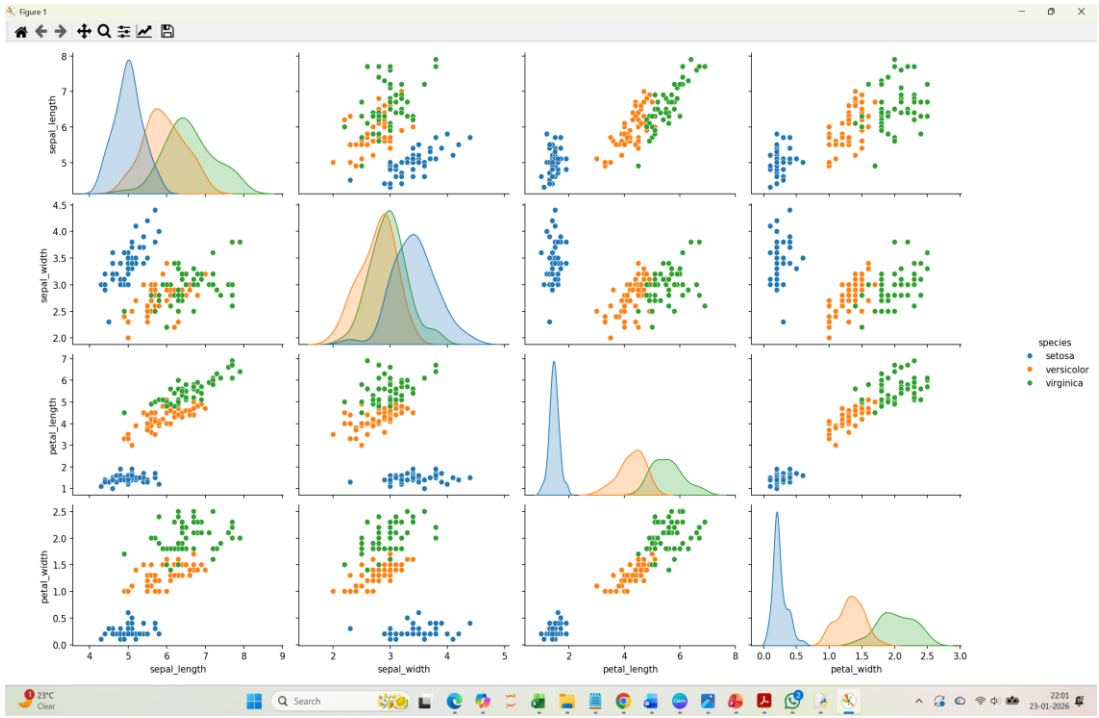
Why Pair plots ?

- Quickly reveals correlations
- Shows non-linear relationships
- Highlights clusters or group separation
- Helps detect outliers
- Provides a fast overview of dataset structure
- Use it when we have 3-10 numerical variables

Python …

```
import seaborn as sns
import matplotlib.pyplot as plt

# Load the iris dataset
iris = sns.load_dataset("iris")

# display a few records
print("Sample data")
print(iris.head())

# Create the pair plot
sns.pairplot(iris, hue="species")

# Show the plot
plt.show()
```

```
Sample data
   sepal_length  sepal_width  petal_length  petal_width species
0           5.1          3.5           1.4          0.2  setosa
1           4.9          3.0           1.4          0.2  setosa
2           4.7          3.2           1.3          0.2  setosa
3           4.6          3.1           1.5          0.2  setosa
4           5.0          3.6           1.4          0.2  setosa
```



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DV Tools – MS Power BI

MS Power BI

- A **Data Visualization** and Business Intelligence tool that converts data from different data sources into interactive dashboards and BI reports.
- **Not a Data Analytics tool**
- Self service tool evolved out of MS Excel & MSBI
- Can connect to several data sources (about 192)
- It can perform Basic Data Wrangling.
- Can generate Visualizations & Dashboards with basic Analytics, Supports Publishing & Sharing.
- Data to Reports in Minutes!!

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Topic 5: Demos

Summary

- Discussed Aggregation by Category.
- Understood grouping mechanisms
- Studied significance of Pivot tables and their types.
- Understood the concept of rolling statistics.
- Understood Feature engineering pipeline and various techniques used in Data Preprocessing.
- Studied a few implementation demos.

Questions?